

Perioscribe

Product & Technical Playbook

Voice-driven AI assistant for periodontal charting,
X-ray analysis, and diagnosis support for UK dental hygienists

Phases 1-2: Product Strategy & Technical Architecture

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Status: Ready for founder review and Phase 3 (implementation) kickoff

Phase 1: Product Strategy

1. Target Audience

Persona 1: "Solo Sarah" -- Independent Hygienist

Attribute	Detail
Role	Self-employed dental hygienist working under direct access (legal since 2013)
Setting	Rents a chair in 2-3 private practices across a city; no dental nurse, no receptionist
Age/Experience	30-45, qualified post-2013 (trained in radiography as part of BSc)
GDC Number	One of ~11,300 registered dental hygienists in the UK (GDC 2025 data)
Hair-on-fire Problem	Simultaneously the clinician, scribe, and admin. During perio assessment she must probe 6 points per tooth (up to 192 measurements), note bleeding, recession, and plaque -- all while hands are in the patient's mouth. She either stops mid-procedure to type/write (breaking sterile field, adding 5-10 min per patient) or tries to remember everything and charts after the patient leaves, risking inaccurate records and medico-legal exposure. At 8-12 patients/day and 5-10 min overhead each, she loses 40-120 minutes daily -- roughly 1-2 billable appointments.
Willingness to Pay	High. Every minute saved is directly recovered revenue. Currently earning GBP 35-55/hr; losing 1 hr/day = GBP 700-1,100/month in lost income. A tool costing GBP 100-200/month is an obvious ROI.
Tech Comfort	Uses iPad or laptop chairside. Comfortable with cloud software (Dentally, SOE).
Decision Authority	100% -- she buys her own tools. No practice manager or IT department.

Persona 2: "Nurse-less Natasha" -- Private Practice Hygienist

Attribute	Detail
Role	Employed hygienist in a private dental practice (1-3 dentists, 1-2 hygienists)
Setting	Dedicated hygiene room; practice has a receptionist but no spare dental nurse for perio exams
Hair-on-fire Problem	Practice principal wants thorough perio documentation for CQC compliance and insurance defence, but will not hire a dedicated dental nurse for the hygiene room (GBP 22,000-28,000/yr salary). Natasha uses a foot pedal and voice recorder, then transcribes after hours. Unpaid admin eats 30-60 min/day. She is frustrated and considering going independent.
Willingness to Pay	The practice owner decides. Pitch: 'Replace a GBP 25,000/yr dental nurse scribe with a GBP 150/month AI tool.'
Decision Authority	Low for Natasha (she requests), high for the practice owner/principal.

Persona 3: "Principal Pete" -- Practice Owner

Attribute	Detail
Role	Owner/principal dentist of a 2-4 surgery private practice
Setting	Employs 2-3 hygienists, 1-2 dentists, dental nurses, reception staff
Hair-on-fire Problem	Staffing is his biggest cost and headache. UK dental nurse vacancy rates at historic highs. Cannot find or afford nurses to scribe during hygiene appointments. CQC and indemnity providers demand better clinical documentation. Hygienists produce inconsistent perio records. Wants standardised, audit-ready documentation without hiring more staff. Also interested in X-ray analysis as a second-opinion tool.
Willingness to Pay	Medium-high. If 2 hygienists each save 1 hr/day = 10 additional appointments/week at GBP 65-85 each = GBP 2,600-3,400/month recovered revenue. Tool at GBP 200-400/month for whole practice is a no-brainer.
Decision Authority	Full. Signs the cheques and chooses the software.

2. Competitive Landscape

Direct Competitors and Adjacent Players

Bola AI (US)

- **What it does:** Voice-powered dental charting and clinical documentation. 'Hey Bola' wake word. Voice Perio, Voice Restorative, AI Scribe. Claims 99% accuracy, 10,000+ users.
- **Pricing:** Custom-quoted, est. USD 129-500+/mo
- **UK presence:** None. US-focused, US PMS only. No UK notation. No GDPR docs.
- **X-ray analysis:** No

Denti.AI (US/Canada)

- **What it does:** Most complete dental AI platform -- FDA-cleared X-ray analysis (Detect), voice perio charting, AI scribe, auto-chart, AI receptionist. 1M+ charts completed.
- **Pricing:** USD 49/mo/location (Detect) + USD 10/user/mo (Voice Perio). Bundle: USD 399/mo.
- **UK presence:** Minimal. FDA-cleared, no MHRA. No UK PMS integrations.
- **X-ray analysis:** Yes (FDA-cleared)

Pearl AI (US)

- **What it does:** FDA-cleared dental X-ray AI ('Second Opinion'). Detects dozens of pathologies. 95%+ accuracy. Also Pearl Voice + Practice Intelligence.
- **Pricing:** Custom-quoted. Enterprise partnerships.
- **UK presence:** Limited. Primarily US. Not MHRA-registered.
- **X-ray analysis:** Yes (core product)

Overjet (US, expanding UK)

- **What it does:** FDA-cleared X-ray AI. Caries detection, bone level measurement, calculus. DentalBee voice docs. Feb 2026: partnered with mydentist (UK's largest DSO).
- **Pricing:** USD 1,000-3,000/mo. Enterprise DSO pricing.
- **UK presence:** YES -- largest dental AI rollout in UK dentistry (mydentist). Has overjet-ai.uk.
- **X-ray analysis:** Yes (core product)

Dentistry Dashboard (UK)

- **What it does:** UK-built AI dental scribe. Clinical notes, referral letters, consent forms, treatment plans from voice. UK dental notation. Registered as medical device (Rule 11).
- **Pricing:** GBP 40/mo (AI Notes), GBP 93/mo (Practice Hub), GBP 156/mo (Growth)
- **UK presence:** Strong. NHS DSPT compliant, Cyber Essentials. Integrates SOE, R4, Carestream, Dentally, Exact.
- **X-ray analysis:** No

Dentally (UK)

- **What it does:** Cloud-based PMS. Patient records, scheduling, NHS claims, imaging integration. Dominant modern PMS in UK private practice.
- **Pricing:** From GBP 79/mo (single) to GBP 149/mo (multi)
- **UK presence:** Market leader for UK cloud PMS. Deep NHS integration.
- **X-ray analysis:** No (but has imaging viewer)

Competitive Gap Analysis

Feature	Bola AI	Denti.AI	Pearl	Overjet	Dent. Dashboard	Dentally	Our Product
Voice perio charting (real-time)	Yes	Yes	Partial	Partial	Partial	No	Yes
X-ray AI analysis	No	Yes (FDA)	Yes (FDA)	Yes (FDA)	No	No	Yes
Combined voice + X-ray + diagnosis	No	Partial	No	No	No	No	Yes
UK dental notation (Palmer)	No	No	No	Partial	Yes	Yes	Yes
UK PMS integration	No	No	No	Partial	Yes	N/A	Yes (Dentally 1st)
UK GDPR / MHRA compliance	No	No	No	Partial	Yes	Yes	Yes
Hygienist-specific UX	No	Partial	No	No	No	No	Yes
BPE scoring	No	No	No	No	Partial	Yes	Yes
Treatment plan generation	Partial	Partial	No	No	Yes	Yes	Yes
NHS integration potential	No	No	No	Partial	Yes	Yes	Future
Offline capability	No	No	No	No	No	No	Yes (basic)
Price (monthly, per user)	\$129-500	\$49-399	Custom	\$1K-3K	GBP 40-156	GBP 79-149	GBP 79-149

Key Strategic Insight

No product on the market combines all three pillars: real-time voice perio charting + X-ray AI analysis + suggested diagnoses, built specifically for UK dental hygienists. The US market has Bola, Denti.AI, Pearl, and DentScribe competing fiercely, but none have meaningful UK presence, compliance, or PMS integrations. Overjet is entering via mydentist (enterprise DSO), leaving the independent/small-practice segment wide open. Dentistry Dashboard is the closest UK competitor but lacks dedicated perio charting and X-ray analysis. The opportunity is to build the 'Dentistry Dashboard for hygienists' -- UK-native, voice-first, with X-ray AI that no UK product currently offers.

3. Lean MVP Scope

Must-Have Features (Weeks 1-8)

#	Feature	Rationale
1	Voice-to-periodontal-chart	Core differentiator. Hygienist speaks pocket depths, system fills a visual 6-point chart in real time. Must handle: 6-point probing depths, BOP, recession, BPE scores. Palmer notation.
2	Patient record management	Create/search patients, store perio history over time, track appointments. Basic demographics (name, DOB, NHS number optional).
3	Perio chart PDF export	Clean, printable perio chart attachable to any PMS record. BPE summary, pocket depth visualisation, bleeding points, recession. Minimum viable integration.
4	X-ray upload + AI analysis	Upload bitewing/periapical images. AI analyses for bone loss, calculus, caries, periapical radiolucencies. Overlay findings with confidence scores.
5	Combined diagnostic summary	Merge perio + X-ray findings. Highlight risk factors. Suggest AAP/EFP periodontal classification (Stage I-IV, Grade A-C).
6	Treatment plan generation	Auto-generate treatment plan based on findings: SRP for pockets > 4mm, referral for > 6mm, recall interval. Editable before saving.
7	GDPR compliance (day 1)	Explicit consent capture. Audio processed and discarded (never stored). Encryption at rest and in transit. UK data residency. DPIA documented.
8	User auth and practice setup	Email/password registration. Practice profile (name, GDC number, address). Single- and multi-practitioner modes.
9	Tablet-optimised responsive UI	iPad-first design. Large touch targets, landscape orientation, glove-friendly. Works on desktop too.

Future Roadmap (Post-MVP)

Phase	Feature	Timeline
Phase 2 (Mo 3-4)	Dentally API integration	4-6 weeks
Phase 2	SOE/R4 integration	6-8 weeks
Phase 2	Plaque index scoring (O'Leary)	2 weeks
Phase 3 (Mo 5-6)	DICOM viewer	4-6 weeks
Phase 3	Patient-facing reports	2-3 weeks
Phase 3	Multi-practice support	3-4 weeks
Phase 4 (Mo 7-9)	Referral letter generation	2-3 weeks
Phase 4	NHS BSA integration (FP17 claims)	8-12 weeks
Phase 4	Clinical photography integration	3-4 weeks
Phase 5 (Mo 10-12)	MHRA medical device registration	3-6 months
Phase 5	Multi-language support (Welsh)	2-3 weeks
Phase 5	CPD reporting	3-4 weeks
Phase 6+	Real-time 3D CBCT analysis	Research
Phase 6+	Orthodontic charting (IOTN)	4-6 weeks

4. Name and Branding

Naming Candidates

#	Name	Rationale	Domain Likelihood
1	Perioscribe	Combines 'perio' (core workflow) + 'scribe' (what it does). Clear, professional, dental-specific. Easy to say: 'I use Perioscribe.'	perioscribe.co.uk / .com likely available
2	Chartly	Short, modern, action-oriented. Friendly and approachable. Less clinical, more SaaS.	chartly.co.uk likely available
3	VoxPerio	'Vox' (voice) + 'Perio'. Latin root gives clinical gravitas. Distinctive.	voxperio.co.uk / .com likely available
4	ProbeAI	'Probe' is the core clinical instrument. Direct and memorable.	probeai.co.uk likely available
5	Gingiva	Anatomical term for gum tissue. Elegant, clinical, distinctive.	gingiva.co.uk possibly taken, gingiva.ai likely available

Recommendation

Perioscribe is the strongest choice. It immediately communicates both the clinical domain (periodontics) and the core function (scribing/documentation). It is professional enough for NHS contexts, memorable enough for word-of-mouth, and specific enough to avoid trademark conflicts. The '-scribe' suffix positions it alongside established products (DentScribe) without copying them.

Runner-up: Chartly -- if a more modern, SaaS-friendly brand is preferred that could expand beyond perio into general dental charting.

Before committing, verify domain availability and run a UK trademark search via the IPO (Intellectual Property Office) database.

Phase 2: Technical Architecture

5. Stack Recommendation

Critical Constraints

This is not a standard SaaS build. Three factors dominate the stack decision:

- **UK data residency** -- patient health data (special category under UK GDPR) must remain in the UK or EU. No US data processing for patient audio or records.
- **Real-time voice processing** -- voice-to-chart must feel instant (< 500ms latency). Rules out batch-only APIs.
- **Medical device regulation risk** -- the AI diagnosis feature may classify as Class IIa. Stack must support auditability, version control, and clinical safety documentation.

Recommended Stack

Layer	Technology	Justification
Frontend	Next.js 16 (App Router) + React 19	Tablet-first PWA. Server components for patient data. PWA for offline basic access.
UI Framework	Tailwind CSS v4 + shadcn/ui + custom dental chart SVG/Canvas	Rapid development. Touch-optimised. Custom perio chart component (no off-the-shelf library exists).
Voice Processing	Deepgram Nova-3 Medical (primary) + Web Speech API (fallback)	Real-time streaming WebSocket, medical vocabulary, accent-agnostic, EU endpoints. USD 0.0077/min.
X-ray Analysis	GPT-4o Vision via Azure OpenAI (UK South)	No FDA/MHRA-cleared API available for 3rd-party integration. Domain-specific prompting + confidence calibration.
Backend / API	Next.js API Routes + Server Actions	Simple stack. Server-side rendering for patient data. No separate backend at MVP scale.
Database	Supabase (London region, AWS eu-west-2)	PostgreSQL with RLS, auth, storage, real-time. London region for UK data residency. GBP 25/mo Pro plan.
Auth	Supabase Auth (email/password + magic link)	Simple, secure. Magic link for clinicians who forget passwords.
Storage	Supabase Storage (London region)	X-ray images stored in UK. Encrypted at rest. RLS per-practice access. Pro includes 100GB.
Payments	Stripe (UK entity)	GBP billing. Direct debit and card. 1.5% + 20p for UK cards.
Hosting	Vercel (London region for functions)	Edge functions in lhr1 (London). Static assets on CDN.
AI / LLM	Azure OpenAI (UK South) -- gpt-4o-mini for text, gpt-4o for vision	UK data residency guaranteed. DSPT compliance.
Monitoring	Sentry (EU region) + Vercel Analytics	Error tracking with EU data residency.

Voice API Deep Comparison

Criterion	Deepgram Nova-3 Medical	OpenAI gpt-4o-mini-transcribe	Google Cloud STT	Web Speech API
Streaming support	Yes (WebSocket)	Limited (Realtime API only)	Yes (gRPC)	Yes (browser-native)
Medical vocabulary	Yes (Medical model)	No custom vocabulary	Custom model training	No customisation
Background noise	Excellent	Good	Good	Poor
Latency	< 300ms	~500ms	< 400ms	< 200ms (local)
Medical accuracy	95%+	~90% general	~92% with custom	~85% (varies)
UK/EU data processing	Yes (EU endpoints)	No (US processing)	Yes (europe-west2)	Yes (local)
Pricing (per hour)	USD 0.46	USD 1.02	USD 0.96	Free
Keyword boosting	Yes	No	Yes	No
HIPAA/GDPR	Yes	BAA available	Yes	N/A

Recommendation: Deepgram Nova-3 Medical as primary. Best combination of streaming support, medical vocabulary, noise handling, and price. Use keyword boosting for dental-specific terms. EU endpoint ensures audio data stays in EU/UK. Fallback: Web Speech API for basic offline mode.

X-ray Analysis Approach Comparison

Approach	Pros	Cons	Recommendation
GPT-4o Vision (OpenAI)	Fastest to implement. Cheap (~USD 0.01-0.05/image).	Not trained on dental images. 41-69% accuracy. Zero sensitivity for caries in studies. Data leaves UK.	Use for MVP with heavy caveats.
Azure OpenAI (UK South)	Same GPT-4o but UK data residency guaranteed. DSPT compliance.	Same accuracy limitations.	Preferred for production.
Pearl / Overjet API	FDA-cleared, 95%+ accuracy, millions of training images.	No public API. Products only, not platform services.	Future phase -- approach for partnership.
Open-source dental AI	Full control, UK-hosted, no API costs.	Requires ML expertise. Models need validation. Higher regulatory burden.	Phase 3-4 option.
Fine-tuned model on Azure	GPT-4o base + dental fine-tuning. UK residency. Better accuracy.	Requires labelled training data. Expensive and complex.	Phase 2-3 with beta user annotations.

6. Data Schema

Entity Relationships

The schema is built around the core clinical workflow: Practice > Practitioner > Patient > Appointment > Voice Session > Periodontal Chart / X-ray.

```

Practice (1) ----< (M) Practitioner
Practice (1) ----< (M) Patient
Practitioner (1) ----< (M) Appointment
Patient (1) ----< (M) Appointment
Patient (1) ----< (M) PeriodontalChart
Patient (1) ----< (M) XrayImage
Patient (1) ----< (M) TreatmentPlan
Patient (1) ----< (M) ClinicalNote
PeriodontalChart (1) ----< (M) ToothReading
PeriodontalChart (1) ---- (1) BPEScore
XrayImage (1) ---- (1) XrayAnalysis
VoiceSession (1) ---- (0..1) PeriodontalChart
Appointment (1) ---- (0..1) VoiceSession
TreatmentPlan (M) ----< (M) TreatmentPlanItem
Practice (1) ----< (M) Subscription
  
```

Key Tables

Table	Key Fields
Practice	id (UUID PK), name, address fields, postcode, phone, email, gdc_practice_number, timestamps
Practitioner	id (UUID PK), practice_id (FK), auth_user_id, name, email, gdc_number, role (hygienist/therapist/dentist/practice_owner), is_active, timestamps
Patient	id (UUID PK), practice_id (FK), name, DOB, nhs_number, contact info, medical_history_notes, consent fields, timestamps
PeriodontalChart	id (UUID PK), patient_id, practitioner_id, appointment_id, voice_session_id, chart_date, BPE scores (6 sextants), plaque/bleeding scores, perio classification (Stage/Grade/Extent), status, timestamps
ToothReading	id (UUID PK), periodontal_chart_id (FK), tooth_number (1-8 Palmer), quadrant (1-4), is_present, 6-point pocket depths, 6-point BOP, 6-point recession, furcation/mobility grades, plaque, suppuration
XrayImage	id (UUID PK), patient_id, practitioner_id, practice_id, image_type (bitewing/periapical/panoramic/cbct), file_path (Supabase Storage), file_size, file_format, tooth_region, timestamps
XrayAnalysis	id (UUID PK), xray_image_id (FK, unique), analysis_model, model_version, findings (JSONB), raw_response, confidence_overall, processing_time_ms, reviewed_by, timestamps
VoiceSession	id (UUID PK), patient_id, practitioner_id, appointment_id, started/ended_at, duration, transcript, extracted_data (JSONB), processing_status, cost tracking
TreatmentPlan	id (UUID PK), patient_id, practitioner_id, perio_chart_id, xray_analysis_id, title, summary, perio_diagnosis, recall_interval, status (draft/approved/in_progress/complete), timestamps
Subscription	id (UUID PK), practice_id (FK), stripe IDs, plan (solo/practice/enterprise), status (trialing/active/past_due/cancelled/paused), trial/period dates
PatientConsent	id (UUID PK), patient_id, consent_type (voice_recording/xray_ai/data_storage/data_sharing), granted, timestamps, ip_address, method, practitioner_id -- GDPR audit trail

Enums Summary

Enum	Values
practitioner_role	hygienist, therapist, dentist, practice_owner
xray_type	bitewing, periapical, panoramic, cbct, other
appointment_type	perio_assessment, scale_and_polish, review, debridement, initial_exam, other
chart_status	in_progress, complete, reviewed
treatment_priority	urgent, soon, routine, monitor
plan_tier	solo, practice, enterprise
consent_type	voice_recording, xray_ai_analysis, data_storage, data_sharing
perio_stage	I, II, III, IV
perio_grade	A, B, C
quadrant (Palmer)	1 (UR), 2 (UL), 3 (LL), 4 (LR)

7. Core User Flows

Flow 1: Voice Periodontal Charting

Precondition: Hygienist is logged in, has selected or created a patient, consent for voice recording captured.

- **Step 1:** Hygienist navigates to patient record, taps 'New Perio Chart'
- **Step 2:** System displays empty 6-point chart (32 teeth x 6 sites, Palmer notation) with dental arch diagram
- **Step 3:** Hygienist taps 'Start Listening' (large, glove-friendly button)
- **Step 4:** System opens WebSocket to Deepgram Nova-3 Medical, streams microphone audio in real-time
- **Step 5:** Hygienist speaks naturally: 'Upper right six: 3, 2, 4, 5, 3, 2' -- chart cells fill in real-time with colour coding (Green 1-3mm, Amber 4-5mm, Red 6mm+)
- **Step 6:** 'Bleeding mesial buccal and distal lingual' -- system marks BOP dots on appropriate sites
- **Step 7:** 'Recession buccal two millimetres' -- system records recession, updates CAL display
- **Step 8:** Hygienist moves to next tooth, process repeats. 'Missing' marks tooth absent
- **Step 9:** Hygienist says 'Stop listening' or taps stop button
- **Step 10:** System presents summary: completed teeth count, aggregate stats (mean PD, % BOP), highlighted deep pockets, auto-calculated BPE scores
- **Step 11:** Hygienist reviews, makes manual corrections (tap any cell to edit), then 'Save Chart'

Flow 2: X-ray Upload and AI Analysis

Precondition: Patient exists, consent for AI analysis captured.

- **Step 1:** Hygienist navigates to patient record, taps 'X-rays' tab, then 'Upload X-ray'
- **Step 2:** File picker accepts JPEG/PNG (MVP), max 50MB. Required metadata: image type, date taken
- **Step 3:** Image uploads to Supabase Storage (UK London region) with progress bar
- **Step 4:** Hygienist taps 'Analyse with AI' -- disclaimer modal confirms AI is a decision-support tool, not a diagnosis
- **Step 5:** System sends image to Azure OpenAI (UK South) via GPT-4o Vision with structured prompt
- **Step 6:** Results displayed: original X-ray with annotation overlays, findings list with type/location/severity/confidence, colour-coded confidence (Green >80%, Amber 60-80%, Red <60%)
- **Step 7:** Hygienist can accept/reject individual findings, add observations, link to perio chart teeth
- **Step 8:** Results saved with full audit trail (model version, raw response, reviewer ID/timestamp)

Flow 3: Combined Diagnosis and Treatment Plan

Precondition: Patient has a completed perio chart and at least one analysed X-ray.

- **Step 1:** Hygienist taps 'Generate Treatment Plan'; system aggregates latest perio chart, X-ray findings, and medical history
- **Step 2:** AI generates combined assessment: periodontal classification (Stage/Grade/Extent), risk factors, key findings summary
- **Step 3:** AI proposes treatment plan: Phase 1 (OHI), Phase 2 (SRP schedule), Phase 3 (review at 6-8 weeks), Phase 4 (referral if needed), recall interval
- **Step 4:** Hygienist reviews, modifies, reorders, adds/removes items, then 'Approve Plan'
- **Step 5:** Export as PDF: clinical version (for records) and patient-friendly version (simplified language)

8. Non-Functional Requirements

8.1 UK GDPR Compliance (Health Data = Special Category)

Requirement	Implementation
Lawful basis	Explicit consent (Art 9(2)(a)) for voice recording and AI analysis. Legitimate interests (Art 6(1)(f)) for basic patient record management.
Consent management	Per-patient, per-purpose consent before any voice/X-ray AI processing. Granular: can consent to charting but decline X-ray AI. Timestamped with method and practitioner ID.
Data minimisation	Audio streamed and discarded -- NEVER stored. Only transcript and structured data retained. X-rays stored only for clinical record.
Data residency	All patient data in Supabase London (AWS eu-west-2). AI via Azure OpenAI UK South and Deepgram EU. No patient data transits US servers.
DPIA	Required before launch. Covers: voice health data processing, AI medical image analysis, storage/access patterns, third-party processors (Deepgram, Azure).
DPO	Required once exceeding ~50 practices (large-scale health data). Privacy lead sufficient at MVP.
Data subject rights	Access (export as PDF/JSON), erasure, portability, withdraw consent (stop AI, retain base clinical record).
Breach notification	ICO notification within 72 hours. Incident response plan documented.
DPAs	Required with Supabase, Deepgram, Microsoft (Azure OpenAI), Stripe.

8.2 Medical Device Regulation (MHRA)

Feature	Medical Device?	Classification	Rationale
Voice-to-chart (transcription)	No	N/A	Pure data capture tool. No clinical decision-making.
Perio chart auto-filling	No	N/A	Data structuring/visualisation. Measurements from clinician's probe.
BPE auto-calculation	Borderline	Class I (if at all)	Arithmetic calculation from user-entered data.
X-ray AI analysis with findings	YES	Class IIa	Analyses medical image and identifies pathological findings.
Periodontal staging/grading	Borderline to YES	Class IIa	Automated disease severity classification. Risk depends on marketing language.
Treatment plan generation	Borderline	Class I or IIa	Lower risk if 'suggested' with practitioner approval. Higher if auto-generated.

MVP Regulatory Strategy:

- Launch voice-to-chart and perio charting as non-device features (data capture/structuring).
- Launch X-ray analysis as a 'decision support tool' with heavy disclaimers. Never present as 'AI diagnosis'.
- Begin MHRA pre-submission dialogue at Month 6 via MHRA Innovation Office (free consultation).
- Budget GBP 15,000-50,000 for Class IIa registration if required (Phase 5 cost, not MVP blocker).

8.3 Performance Requirements

Metric	Target	Rationale
Voice-to-chart latency	< 500ms	Must feel 'real-time' or hygienist reverts to manual charting
X-ray analysis time	< 15 seconds per image	Hygienist will not wait longer; can continue other tasks while processing

Metric	Target	Rationale
Page load time (tablet)	< 2 seconds on 4G	Many UK practices have mediocre broadband; mobile 4G fallback common
Uptime	99.5%	Clinical tool -- downtime during appointments erodes trust
Voice recognition accuracy	> 95% (dental terms)	Below 95%, manual corrections negate the time savings
Concurrent voice sessions	50+	One session per hygienist; at scale (50 practices x 1 hygienist)

9. Deployment Plan -- Day 1 Infrastructure

Service Configuration and Costs

Service	Plan	Region	Monthly Cost (GBP)	Notes
Vercel	Pro	London (lhr1)	~16	Server functions pinned to London. Preview deployments.
Supabase	Pro	AWS eu-west-2 (London)	~20	8GB DB, 100GB storage, 250K MAU, unlimited API.
Deepgram	Pay-as-you-go	EU endpoint	Variable	Nova-3 Medical streaming at USD 0.0077/min.
Azure OpenAI	Pay-as-you-go	UK South	Variable	GPT-4o Vision for X-ray. GPT-4o-mini for text.
Stripe	Standard	UK	1.5% + 20p/txn	Processing fees only. No monthly fee.
Sentry	Team	EU	~20	Error monitoring, performance tracking.
Domain	Registration	N/A	~1	.co.uk via Cloudflare.
Cloudflare	Free	Global	0	DNS, DDoS, SSL.
SendGrid/Resend	Free tier	N/A	0	Transactional emails.

Total Monthly Cost by Scale

Scale	Fixed Costs	Deepgram	Azure OpenAI	Stripe Fees	Total
10 practitioners (launch)	GBP 57	GBP 55	GBP 16	~GBP 12	~GBP 140/mo
50 practitioners (month 6)	GBP 57	GBP 275	GBP 80	~GBP 60	~GBP 472/mo
100 practitioners (month 12)	GBP 57	GBP 550	GBP 160	~GBP 120	~GBP 887/mo

Revenue vs. Cost at Each Scale

Scale	Monthly Revenue (est.)	Monthly Cost	Gross Margin
10 practitioners @ GBP 99/mo avg	GBP 990	GBP 140	86%
50 practitioners @ GBP 109/mo avg	GBP 5,450	GBP 472	91%
100 practitioners @ GBP 119/mo avg	GBP 11,900	GBP 887	93%

Pricing Model

Plan	Price (GBP/mo)	Target Persona	Includes
Solo	GBP 79	Independent hygienist (Persona 1)	1 practitioner, unlimited patients, voice charting, basic X-ray analysis (20/mo), PDF export
Practice	GBP 149	Small practice (Personas 2-3)	Up to 5 practitioners, unlimited patients, voice charting, X-ray analysis (100/mo), treatment plans, PMS integration
Enterprise	Custom	Multi-site practices, DSOs	Unlimited practitioners, unlimited X-rays, dedicated support, custom integrations, MHRA compliance support

- **Per-patient session fee:** GBP 0.50 per voice charting session (covers Deepgram cost ~GBP 0.04/session with healthy margin)
- **Free trial:** 14 days, all features, no credit card required. Beta testers get 6 months free.

Appendix A: Regulatory Risk Matrix

Risk	Likelihood	Impact	Mitigation
MHRA classifies X-ray AI as Class IIa	High	High	Frame as 'decision support,' not diagnosis. Engage MHRA Innovation Office at month 6. Budget GBP 15-50K for registration.
ICO investigation for GDPR non-compliance	Low (if DPIA done)	Very High	Complete DPIA before launch. Appoint privacy lead. DPAs with all processors.
Voice recognition fails in noisy clinical environment	Medium	High	Use Deepgram Medical (noise-trained). Keyword boosting. Extensive testing in real clinical settings with beta testers.
GPT-4o Vision produces incorrect X-ray findings	High	Very High	Never present as diagnosis. Confidence scores on all findings. Mandatory practitioner review. Disclaimer on every result.
Competitor enters UK small-practice market	Medium	Medium	Move fast. UK-native compliance and PMS integration are moats. Build community not just software.
Dental hygienist adoption is slow	Medium	High	Leverage beta testers as champions. Target BSDHT conference. Referral incentives.
UK data residency requirements tighten	Low	Medium	Architecture designed for UK residency. Azure UK South available as fallback for all services.

Appendix B: Key Regulatory References

- **ICO -- Special category data guidance:** ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/lawful-basis/special-category-data/
- **ICO -- DPIA guidance:** ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/accountability-and-governance/data-protection-impact-assessments-dpias/
- **MHRA -- Software and AI as a medical device:** gov.uk/government/publications/software-and-artificial-intelligence-ai-as-a-medical-device
- **GDC -- Scope of Practice:** gdc-uk.org/docs/default-source/scope-of-practice/scope-of-practice.pdf
- **NHS England -- DCB0129 clinical risk management:** digital.nhs.uk/data-and-information/information-standards/governance/latest-activity/standards-and-collections/dcb0129
- **Data (Use and Access) Act 2026:** Royal Assent 19 June 2026, all data protection provisions now in force

Day 1 Checklist

- Domain registered (perioscribe.co.uk + perioscribe.com)
- Supabase project created (London region)
- Vercel project created (functions pinned to lhr1)
- Azure OpenAI resource created (UK South)
- Deepgram account created (EU endpoint configured)
- Stripe account created (UK entity, GBP)
- Sentry project created (EU)
- DNS configured (Cloudflare)
- SSL certificates provisioned (automatic via Vercel)
- Environment variables set (production + preview)
- GDPR Privacy Policy drafted
- Cookie Policy drafted
- Terms of Service drafted
- DPIA template started
- Patient consent flow designed

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